

EXPERIMENT 6

TITLE

Vacuum Cleaner Problem

AIM

To write a Python program to simulate the **Vacuum Cleaner Problem**.

OBJECTIVE

- To understand the working of a simple reflex agent.
- To clean all dirty rooms.
- To demonstrate intelligent agent behavior.

ALGORITHM

Step 1: Define two rooms.

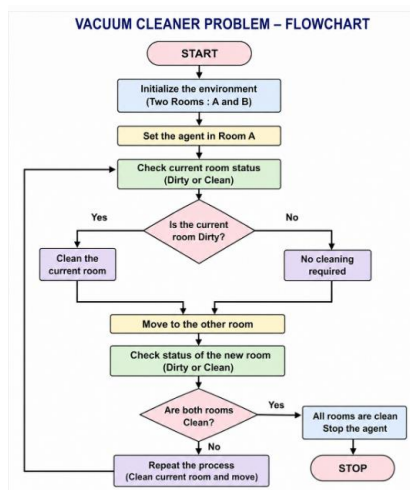
Step 2: Check the current room.

Step 3: If the room is dirty, clean it.

Step 4: Move to the next room.

Step 5: Repeat until all rooms are clean.

FLOWCHART



PYTHON CODE

```
rooms = {
    "A": "Dirty",
    "B": "Dirty"
}

for room in rooms:

    print("Checking Room", room)

    if rooms[room] == "Dirty":

        print("Cleaning Room", room)

        rooms[room] = "Clean"

    else:

        print("Already Clean")

print("\nFinal Status")

for room in rooms:
    print(room, "=", rooms[room])
```

SAMPLE OUTPUT

```
Checking Room A
Cleaning Room A
```

```
Checking Room B
Cleaning Room B
```

```
Final Status
```

```
A = Clean
B = Clean
```

CONCLUSION

The Vacuum Cleaner Problem was implemented successfully. The agent sensed the environment, cleaned the dirty rooms, and achieved the goal of making all rooms clean.